

There are a total of five problems. You have to solve all of them.

Problem 1 (CO3): Designing Context-Free Grammars (10 points)

Let $\Sigma = \{0, 1\}$. Consider the following pair of languages. Recall that for a string w , $|w|$ denotes the length of w .

$$L_1 = \{w \in \Sigma^* : w \text{ contains exactly two 1s}\}$$

$$L_2 = \{x\#y : x \in \Sigma^*, y \in L_1, |x| = |y|\}$$

Now solve the following problems.

- (a) Give a context-free grammar for the language L_1 . (4 points)
- (b) Write down four seven-letter strings in L_2 . (1 point)
- (c) Give a context-free grammar for the language L_2 . (5 points)

Problem 2 (CO3): Derivations, Parse Trees and Ambiguity (10 points)

Take a look at the grammar below and solve the following problems.

$$S \rightarrow AB \mid \mathbf{b}AA \mid SS$$

$$A \rightarrow \mathbf{a} \mid \mathbf{a}A$$

$$B \rightarrow \mathbf{b} \mid \epsilon$$

- (a) Give a leftmost derivation for the string $\mathbf{abaaab}$. (3 points)
- (b) Sketch the parse tree corresponding to the derivation you gave in (a). (2 points)
- (c) Demonstrate that the given grammar is ambiguous by showing two more parse tree (apart from the one you already found in (b)) for the same string. (4 points)
- (d) Find a string w of length five such that w has exactly one parse tree in the grammar above. (1 point)

Problem 3 (CO4): The CYK Algorithm (10 points)

Apply the CYK algorithm to determine whether the string $\mathbf{yxzyz}$ can be derived in the following grammar. You must show the entire CYK table. Here $\mathbf{x}$, $\mathbf{y}$ and $\mathbf{z}$ are terminals, and the rest are variables.

$$S \rightarrow AB$$

$$A \rightarrow AC \mid \mathbf{y}$$

$$B \rightarrow BD \mid EB \mid \mathbf{z}$$

$$C \rightarrow AE \mid \mathbf{x}$$

$$D \rightarrow \mathbf{y}$$

$$E \rightarrow BD \mid \mathbf{x}$$

Problem 4 (CO4): Chomsky Normal Form (10 points)

Answer the following questions.

- (a) List the productions that violate the conditions of the Chomsky Normal Form (CNF) in the following grammar. (4 points)

$$\begin{aligned} S &\rightarrow a \mid AB \\ A &\rightarrow SB \mid ba \mid B \\ B &\rightarrow b \mid \epsilon \end{aligned}$$

- (b) Write down the additional rules that need to be added to the following grammar if the production $X \rightarrow \epsilon$ is removed. (4 points)

$$\begin{aligned} S &\rightarrow aXbXcX \mid X \\ X &\rightarrow a \mid \epsilon \end{aligned}$$

- (c) Write down the additional rules that need to be added to the following grammar if all the unit productions are removed. (2 points)

$$\begin{aligned} S &\rightarrow Aa \\ A &\rightarrow BC \mid C \\ B &\rightarrow b \\ C &\rightarrow A \mid AA \mid AaB \end{aligned}$$

Problem 5 (CO3): Constructing Pushdown Automata (10 points)

Let $\Sigma = \{0, 1\}$. Note that we define $c(w, x)$ to be the count of x in the string w .

$$\begin{aligned} L_1 &= \{w \mid w \text{ is a palindrome and the length of } w \text{ is odd.}\} \\ L_2 &= \{w_1\#w_2\#w_3 \mid c(w_1, 0) = c(w_2, 0) \text{ or } c(w_2, 1) = c(w_3, 1)\} \end{aligned}$$

Now solve the following problems.

- (a) Give the state diagram of a pushdown automaton that recognizes L_1 . (4 points)
(b) Find all strings $w \in L_2$ such that w starts with 00010#1001# and has a length of 14. (1 point)
(c) Give the state diagram of a pushdown automaton that recognizes L_2 . (5 points)

There are a total of five problems. You have to solve all of them.

Problem 1 (CO3): Designing Context-Free Grammars (10 points)

Let $\Sigma = \{0, 1\}$. Consider the following pair of languages. Recall that for a string w , $|w|$ denotes the length of w .

$$L_1 = \{w \in \Sigma^* : w \text{ contains exactly two 0s}\}$$

$$L_2 = \{x\#y : x \in L_1, y \in \Sigma^*, |x| = |y|\}$$

Now solve the following problems.

- (a) Give a context-free grammar for the language L_1 . (4 points)
- (b) Write down four seven-letter strings in L_2 . (1 point)
- (c) Give a context-free grammar for the language L_2 . (5 points)

Problem 2 (CO3): Derivations, Parse Trees and Ambiguity (10 points)

Take a look at the grammar below and solve the following problems.

$$S \rightarrow BA \mid Abb \mid SS$$

$$A \rightarrow a \mid \epsilon$$

$$B \rightarrow b$$

- (a) Give a leftmost derivation for the string babbbba. (3 points)
- (b) Sketch the parse tree corresponding to the derivation you gave in (a). (2 points)
- (c) Demonstrate that the given grammar is ambiguous by showing two more parse tree (apart from the one you already found in (b)) for the same string. (4 points)
- (d) Find a string w of length five such that w has exactly one parse tree in the grammar above. (1 point)

Problem 3 (CO4): The CYK Algorithm (10 points)

Apply the CYK algorithm to determine whether the string yxxzy can be derived in the following grammar. You must show the entire CYK table. Here x , y and z are terminals, and the rest are variables.

$$S \rightarrow AB$$

$$A \rightarrow AC \mid y$$

$$B \rightarrow BD \mid EB \mid z$$

$$C \rightarrow AE \mid x$$

$$D \rightarrow y$$

$$E \rightarrow BD \mid x$$

Problem 4 (CO4): Chomsky Normal Form (10 points)

Answer the following questions.

- (a) List the productions that violate the conditions of the Chomsky Normal Form (CNF) in the following grammar. (4 points)

$$\begin{aligned} S &\rightarrow a \mid AB \\ A &\rightarrow SB \mid ba \mid B \\ B &\rightarrow b \mid \epsilon \end{aligned}$$

- (b) Write down the additional rules that need to be added to the following grammar if the production $X \rightarrow \epsilon$ is removed. (4 points)

$$\begin{aligned} S &\rightarrow aXbXcX \mid X \\ X &\rightarrow a \mid \epsilon \end{aligned}$$

- (c) Write down the additional rules that need to be added to the following grammar if all the unit productions are removed. (2 points)

$$\begin{aligned} S &\rightarrow Aa \\ A &\rightarrow BC \mid C \\ B &\rightarrow b \\ C &\rightarrow A \mid AA \mid AaB \end{aligned}$$

Problem 5 (CO3): Constructing Pushdown Automata (10 points)

Let $\Sigma = \{a, b\}$. Note that we define $c(w, x)$ to be the count of x in the string w .

$$\begin{aligned} L_1 &= \{w \mid w \text{ is a palindrome and the length of } w \text{ is even.}\} \\ L_2 &= \{w_1\#w_2\#w_3 \mid c(w_1, b) = c(w_3, b) \text{ or } c(w_2, a) = c(w_3, a)\} \end{aligned}$$

Now solve the following problems.

- (a) Give the state diagram of a pushdown automaton that recognizes L_1 . (4 points)
(b) Find all strings $w \in L_2$ such that w starts with `babbab#abba#` and has a length of 15. (1 point)
(c) Give the state diagram of a pushdown automaton that recognizes L_2 . (5 points)